Elliptic Curves Over Finite Fields and Applications to Cryptography

2026 School on Arithmetic, Geometry and Algorithms

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Introduction

These lecture notes accompany the lectures for the course on elliptic curves over finite fields and applications to cryptography for the 2026 School on Arithmetic, Geometry and Algorithms. The main references I used are [Kun26, Sil09]. Other useful references are [HPS14, AM26].

At the heart of public-key cryptography are functions that are easy to compute but are believed hard to reverse without secret information. In these notes, we first build the arithmetic of elliptic curves over finite fields from explicit formulas. We then use it to formulate the discrete logarithm problem and elliptic-curve Diffie–Hellman. At the end, we will discuss isogenies and their cryptographic context.

These notes were prepared with the help of ChatGPT 5.6 Terra. All errors are my own.

Explicit computations

The ideal way to follow these notes is to try examples in your favorite computer algebra system. I recommend you use the open source software SageMath, but you are welcome to use whatever you prefer (Magma, Oscar, PARI/GP, etc.).

Prerequisites

The required background includes arithmetic in $\mathbb{F}_p$: modular addition, multiplication, inverses, and polynomial evaluation. Basic group theory is also used in the course.

A note from the author

Despite my best efforts, these notes may contain typos. If you spot any, please feel free to email me; I appreciate your help!

Acknowledgments

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Lecture 1: Elliptic curves over finite fields

This course starts with arithmetic in prime fields. Our immediate goal is to construct a finite abelian group from a polynomial equation and learn to compute in it. The cryptographic applications in later lectures use exactly this arithmetic, but with primes and subgroup orders too large to enumerate.

Throughout this lecture, $p > 3$ is a prime.

1.1 A short toolkit

1.1.1 Arithmetic in $\mathbb{F}_p$

The elements of $\mathbb{F}_p$ are $0, 1, \ldots, p-1$, with addition and multiplication reduced modulo p. Every nonzero element $a \in \mathbb{F}_p$ has a unique inverse a^{-1}, so division by a means multiplication by a^{-1}. For example, in $\mathbb{F}_7$ we have $2^{-1} = 4$, since $2 \cdot 4 = 8 = 1$.

Definition 1.1. An element $r \in \mathbb{F}_p$ is a quadratic residue if $r = y^2$ for some $y \in \mathbb{F}_p$. Otherwise it is a quadratic nonresidue.

For p odd, every nonzero quadratic residue has exactly two square roots, y and $-y$, while 0 has one. Thus a table of squares is enough to count points on the curves in this lecture.

Example 1.2. The squares in $\mathbb{F}_7$ are

$$0^2 = 0, \quad 1^2 = 6^2 = 1, \quad 2^2 = 5^2 = 4, \quad 3^2 = 4^2 = 2.$$

Hence the quadratic residues in $\mathbb{F}_7$ are $0, 1, 2, 4$.

✎ **Exercise 1.3.** In this problem, you will study quadratic residues in finite fields.

1. Find the quadratic residues in $\mathbb{F}_{11}$ and $\mathbb{F}_{13}$. How many are there in each case?
2. Can you find a pattern in the number of quadratic residues in $\mathbb{F}_p$ for odd primes p?
3. Can you prove the formula you found? *Hint: consider the map $x \mapsto x^2$ on the nonzero elements of $\mathbb{F}_p$.*

1.2 Curves and their points

An elliptic curve over a field k is a geometric object described here by an equation in two variables. In this course we use the short Weierstrass equation

$$y^2 = x^3 + ax + b,$$

where a and b are chosen constants in k. Usually, k will ddenote the finite fields $\mathbb{F}_p$ or $\mathbb{F}_{p^2}$. A solution $(x, y) \in k^2$ is called an affine point of the curve, or simply a point of the curve.

For example, on $y^2 = x^3 + x + 1$ over $\mathbb{F}_7$, taking $x = 2$ gives

$$y^2 = 2^3 + 2 + 1 = 4.$$

In 1.2, we saw that 4 is a quadratic residue in $\mathbb{F}_7$, so there are two values of y that satisfy the equation. Since $2^2 = 4$ and $5^2 = 4$ in $\mathbb{F}_7$, this value $x = 2$ gives the two affine points $(2, 2)$ and $(2, 5)$. For a fixed x, there can similarly be zero, one, or two affine points, according as the right-hand side is a nonresidue, zero, or a nonzero quadratic residue.

Finally, not every equation of this shape gives an elliptic curve. We require that it have no singular point—no crossing or cusp in its geometric picture. For a short Weierstrass equation in characteristic different from 2 and 3, this is exactly the nonvanishing condition in the following definition. It is what makes the addition law well behaved.

Definition 1.4. Recall that $p > 3$. An elliptic curve in short Weierstrass form over $\mathbb{F}_p$ is an equation

$$E\colon y^2 = x^3 + ax + b, \qquad a, b \in \mathbb{F}_p, \tag{1.5}$$

such that

$$4a^3 + 27b^2 \neq 0.$$

Its set of $\mathbb{F}_p$-points is

$$E(\mathbb{F}_p) = \{(x, y) \in \mathbb{F}_p^2 : y^2 = x^3 + ax + b\} \cup \{O\}.$$

The extra symbol O, called the point at infinity, will be the identity element of the group law.

To see where O comes from, homogenize the equation:

$$Y^2 Z = X^3 + aXZ^2 + bZ^3.$$

In projective coordinates the solutions with $Z = 0$ consist of the single point $[0 : 1 : 0]$, represented in the affine formulas by O. Including this point is what allows point addition to have an identity.

Remark 1.6. The restriction $p > 3$ is deliberate: it makes this short equation and the formulas below valid without exceptional characteristic-2 or characteristic-3 cases.

✎ **Exercise 1.7.** Graph the real solutions of $y^2 = f(x)$ for

$$f(x) = x^3 - 1, \qquad f(x) = x^3 - x, \qquad f(x) = x^3 - x^2.$$

Which of these equations are nonsingular and hence define elliptic curves over $\mathbb{R}$?

Definition 1.8. The discriminant of (1.5) is

$$\Delta(E) = -16(4a^3 + 27b^2).$$

The condition $\Delta(E) \neq 0$ says that the cubic on the right has no repeated root. In this case, we say that E is nonsingular. If $\Delta(E) = 0$, then E is singular and has a cusp or crossing point.

✎ **Exercise 1.9.** For which primes $p > 3$ does $y^2 = x^3 + x + 2$ fail to define an elliptic curve over $\mathbb{F}_p$? What about over $\mathbb{F}_{p^2}$?

✎ **Exercise 1.10.** Let k be a field of characteristic different from 2 and 3, and let $f(x) = x^3 + ax + b$.

1. Show that $4a^3 + 27b^2 = 0$ if and only if f has a repeated root.
2. Use the partial derivatives of $g(x, y) = y^2 - x^3 - ax - b$ to show that f having a repeated root is equivalent to the existence of a singular affine point on the curve.

1.2.1 Points on elliptic curves

The same equation can be studied over any field containing its coefficients.

Definition 1.11. Let k and K be fields with $k \subseteq K$ and k being a subfield of K. Then we say that K is an extension field of k. We denote this by K/k.

If E is defined over k and K/k is a field extension, then

$$E(K) := \{(x, y) \in K^2 \,:\, y^2 = x^3 + ax + b\} \cup \{O\}.$$

The notation $\#E(K)$ denotes the number of these points when the set is finite.

Example 1.12 (One curve over several fields)**.** Consider the nonsingular short Weierstrass equation

$$E\colon y^2 = x^3 - 27x + 2970,$$

Over $\mathbb{C}$, every value of x except a root of the cubic gives two values of y, so $E(\mathbb{C})$ is infinite. Over $\mathbb{R}$,

$$x^3 - 27x + 2970 = (x + 15)(x^2 - 15x + 198),$$

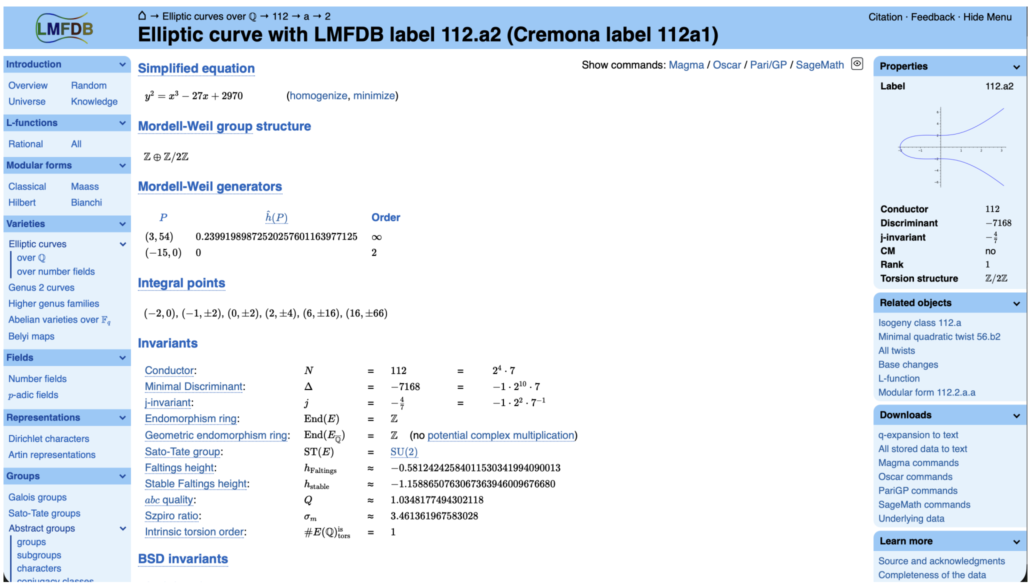
and $x^2 - 15x + 198 > 0$ for every real x. Thus $E(\mathbb{R})$ has affine points exactly when $x \geq -15$, including $(-15, 0)$. Over $\mathbb{Q}$, the points $(-15, 0)$, $(3, \pm 54)$, and $(21, \pm 108)$ are rational.

Over $\mathbb{F}_{11}$, direct enumeration gives

$$E(\mathbb{F}_{11}) = \{O, (0,0), (2, \pm 3), (3, \pm 1), (4, 0), (5, \pm 1), (7, 0), (10, \pm 2)\},$$

so $\#E(\mathbb{F}_{11}) = 12$. Over the quadratic extension $\mathbb{F}_{11^2}$ the same curve has 144 points. The points over $\mathbb{F}_{11}$ are included among these extension-field points. Questions about integral points, obtained by asking for integer coordinates rather than field-valued coordinates, lead to deeper arithmetic geometry; see [Zag87].

A useful source for further examples and arithmetic data is the LMFDB elliptic-curve database. The page for the curve in Example 1.12 is this. It contains a wealth of information, including the number of points over many finite fields, the group structure of $E(\mathbb{F}_p)$ for many primes p, and the torsion subgroup over $\mathbb{Q}$.



> ✎ **Exercise 1.13.** For each curve below, compute $\#E(\mathbb{F}_5)$. Then use SageMath to compute $\#E(\mathbb{F}_{5^2})$. Do you see a pattern?
>
> $$y^2 = x^3 - x, \qquad y^2 = x^3 + x + 1.$$

1.2.2 Counting points by hand

For each $x \in \mathbb{F}_p$, compute $x^3 + ax + b$ and look it up in the table of quadratic residues. A nonzero residue gives two points, 0 gives one, and a nonresidue gives none. Finally add O.

Example 1.14. Let $E/\mathbb{F}_7$ be

$$E\colon y^2 = x^3 + x + 1.$$

The values of $x^3 + x + 1$ for $x = 0, 1, \ldots, 6$ are

$$1, 3, 4, 3, 6, 5, 6.$$

Only 1 and 4 are quadratic residues in this list. Therefore

$$E(\mathbb{F}_7) = \{O, (0,1), (0,6), (2,2), (2,5)\}, \qquad \#E(\mathbb{F}_7) = 5.$$

✎ **Exercise 1.15.** Count the points on each curve over $\mathbb{F}_7$:

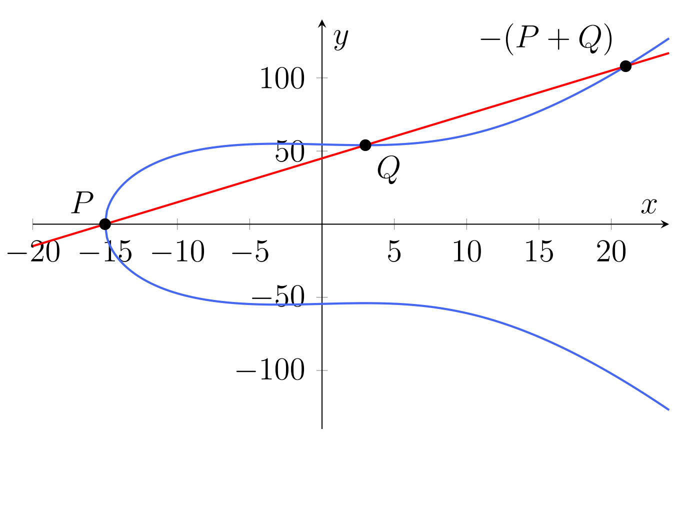
$$y^2 = x^3 - x, \qquad y^2 = x^3 + 3.$$

Check that their point counts are 8 and 13, respectively.

1.3 The group law

Over the real numbers one can picture the group law by drawing a line through two points, finding the third intersection, and reflecting in the x-axis. Over $\mathbb{F}_p$ there is no picture, but the same algebraic formulas work using modular inverses.

Equivalently, three collinear points on a nonsingular cubic add to O (with intersection multiplicities counted). For the real curve in Example 1.12, the line through $P = (-15, 0)$ and $Q = (3, 54)$ also meets the curve at $R = (21, 108)$. Thus $P + Q + R = O$, and $P + Q = (21, -108)$.



Theorem 1.16 (Addition formulas)**.** *Let $E : y^2 = x^3 + ax + b$ be an elliptic curve over $\mathbb{F}_p$.*

1. *O is the identity: $P + O = O + P = P$.*
2. *The inverse of (x, y) is $-(x, y) = (x, -y)$.*
3. *If $P = (x_1, y_1)$ and $Q = (x_2, y_2)$ with $x_1 \neq x_2$, put*
$$m = \frac{y_2 - y_1}{x_2 - x_1}, \qquad x_3 = m^2 - x_1 - x_2, \qquad y_3 = m(x_1 - x_3) - y_1.$$
Then $P + Q = (x_3, y_3)$.
4. *If $P = (x_1, y_1)$ with $y_1 \neq 0$, put*
$$m = \frac{3x_1^2 + a}{2y_1}, \qquad x_3 = m^2 - 2x_1, \qquad y_3 = m(x_1 - x_3) - y_1.$$
Then $[2]P = (x_3, y_3)$.
5. *If $Q = -P$, including the case $y_1 = y_2 = 0$, then $P + Q = O$.*

With these rules, $(E(\mathbb{F}_p), +)$ is a finite abelian group.

The associativity assertion is the hard part of this theorem; it follows from the projective geometry of a nonsingular cubic. A proof of the group law and the formulas appears in [Sil09, Ch. III, §§2–3]. We will use the theorem, not attempt that proof here.

Example 1.17. Continue with Example 1.14. For $P = (0, 1)$, doubling gives
$$m = \frac{1}{2} = 4 \pmod 7, \qquad [2]P = (2, 5).$$
Adding P to $(2, 5)$ gives $[3]P = (2, 2)$, then $[4]P = (0, 6)$, and finally $[5]P = O$. Thus P generates all of $E(\mathbb{F}_7)$.

Definition 1.18. The scalar multiplication $[n]P$ is computed efficiently by double-and-add: write n in binary, repeatedly double the current point, and add only at the binary digits equal to 1. This needs only about $\log_2 n$ doublings and additions.

✎ **Exercise 1.19.** Use double-and-add to compute $[13]P$ for $P = (0, 1)$ on the curve in Example 1.14. Explain why the answer could also have been found by reducing 13 modulo $\mathrm{ord}(P)$.

Remark 1.20. If instead of $\mathbb{F}_p$ we work over the rational numbers $\mathbb{Q}$, then the group $E(\mathbb{Q})$ is finitely generated by Mordell-Weil's theorem. The torsion subgroup is classified by Mazur's theorem, and the rank of the free part is still the subject of current research. It is not even known whether the rank can be arbitrarily large. You can find a list of the largest known ranks at this link.

1.4 How many points can there be?

The elementary bound $\#E(\mathbb{F}_q) \leq 2q+1$ follows by counting the at most two y-values above each x. Hasse's theorem is vastly sharper; see [Sil09, Ch. V, §1].

Theorem 1.21 (Hasse). *If E is an elliptic curve over $\mathbb{F}_q$, then*

$$q+1-2\sqrt{q} \leq \#E(\mathbb{F}_q) \leq q+1+2\sqrt{q}.$$

For $q = 7$, the possible integer values lie between 3 and 13. This explains why the examples above can have dramatically different point counts, even though both are defined over the same field.

Example 1.22. The two ends of Hasse's interval already occur over $\mathbb{F}_7$:

$$E_1 : y^2 = x^3 - 3 \quad \text{has } \#E_1(\mathbb{F}_7) = 3, \qquad E_2 : y^2 = x^3 + 3 \quad \text{has } \#E_2(\mathbb{F}_7) = 13.$$

Thus the variation in point counts is real, even for a very small field. The points are:

$$E_1(\mathbb{F}_7) = \{O, (0,2), (0,5)\},$$
$$E_2(\mathbb{F}_7) = \{O, (1,2), (1,5), (2,2), (2,5), (3,3), (3,4), (4,2), (4,5), (5,3), (5,4), (6,3), (6,4)\}.$$

✎ **Exercise 1.23.** Use your answer to Exercise 1.3 to coumpute a bound for $\#E(\mathbb{F}_q)$ better than the naive bound $\#E(\mathbb{F}_q) \leq 2q+1$.

✎ **Exercise 1.24.** Enumerate the curves $y^2 = x^3 + ax + b$ over $\mathbb{F}_7$ with nonzero discriminant. Verify that the curves in Example 1.22 attain the smallest and largest possible point counts.

✎ **Exercise 1.25.** For $p = 5, 7, 11$, use SageMath to enumerate all pairs (a, b) with $4a^3 + 27b^2 \neq 0$ and record the possible values of $\#E(\mathbb{F}_p)$. Compare your data with Hasse's interval.

✎ **Exercise 1.26.** For several primes p, compute $\lfloor p+1+2\sqrt{p} \rfloor$ and enumerate all nonsingular curves $y^2 = x^3 + ax + b$ over $\mathbb{F}_p$. Let $M(p)$ be the number of pairs (a, b) whose curve has this maximal permitted number of points. Record the values of $M(p)$ and look for a pattern.

1.5 Practice problems

The following problems collect the operations used throughout the lecture. Work the first three by hand before checking any computations with SageMath.

✎ **Exercise 1.27.** Over $\mathbb{F}_{11}$, consider

$$E\colon y^2 = x^3 + 2x + 7.$$

Compute the discriminant and verify that E is nonsingular. Make a table of the quadratic residues in $\mathbb{F}_{11}$, then use it to enumerate $E(\mathbb{F}_{11})$.

✎ **Exercise 1.28.** Use your point count for $E/\mathbb{F}_{11}$ to check Hasse's bound. Determine the group structure of $E(\mathbb{F}_{11})$.

These are the same operations implemented by an elliptic-curve cryptographic library. The only difference in practice is the size of the field and the subgroup: production parameters make direct enumeration and trial search infeasible.

Lecture 2: The elliptic-curve discrete logarithm problem

The group law turns an elliptic curve into a setting for public-key cryptography. In this lecture, we will study the theoretical foundations of the elliptic-curve discrete logarithm problem (ECDLP).

2.1 Public-key cryptography

Public-key cryptography lets people who have never met establish secure communication over a public channel. The channel might be the Internet, a wireless network, or any other medium on which an adversary can read the messages. We will call the two people who want to communicate Alice and Bob, and call an adversary Eve. A cryptographic protocol specifies what Alice and Bob send, what they keep secret, and what Eve is allowed to see.

The basic aim is to turn a secret key into something useful. A secret key is a short piece of information used by a cryptographic algorithm. For example, a symmetric encryption algorithm uses the *same* secret key to encrypt and decrypt:

$$\text{plaintext} \xrightarrow{\text{encrypt with } K} \text{ciphertext} \xrightarrow{\text{decrypt with } K} \text{plaintext}.$$

Symmetric encryption is fast and is what normally protects the contents of a long message. Its difficulty is key distribution: how can Alice and Bob obtain the common secret key K in the first place if Eve can observe their communication?

Public-key cryptography addresses this problem by giving each participant a pair of related keys:

$$\text{public key} \qquad \text{and} \qquad \text{private key}.$$

The public key may be published or sent over the public channel. The private key must be sampled randomly and kept only by its owner. The two keys are constructed so that a public key can be computed efficiently from its private key, while recovering the private key from the public key is believed to be computationally infeasible. Security does not rely on hiding the algorithm: the curve, the protocol, and all public parameters are assumed to be known to Eve.

Public-key techniques have two common roles. First, key agreement allows Alice and Bob to derive a common symmetric key; Diffie–Hellman is the example studied below. Second, digital signatures allow a sender to prove possession of a private key, so that anyone with the corresponding public key can verify the sender's signature. In practice, a protocol often uses public-key cryptography to authenticate participants and agree on a temporary symmetric key, then uses symmetric encryption to protect the data.

For elliptic-curve cryptography, the relevant one-way-looking operation is *scalar multiplication*. Public parameters include an elliptic curve $E/\mathbb{F}_q$ and a publicly known point P.

Alice chooses a secret integer a (her private key) and computes

$$A = [a]P.$$

She may publish A as her public key. Repeatedly adding P to itself (using the fast double-and-add algorithm) computes $[a]P$ efficiently. The reverse task—given P and A, recover a—is the elliptic-curve discrete logarithm problem. The remainder of this lecture makes this statement precise and explains how it leads to Diffie–Hellman key agreement.

Warning 2.1. Public information is not necessarily harmless information. A public key is safe to reveal only because the associated mathematical problem is hard for the chosen parameters. Tiny examples are excellent for learning the protocols, but their private keys can be found by simply trying all possibilities. Moreover, mathematical hardness alone does not provide every security property: key agreement must be authenticated to prevent an active attacker from impersonating a participant.

2.2 Further structure of the points on elliptic curves

Before we start looking at the discrete logarithm problem, we need to understand the structure of the group of points on an elliptic curve over a finite field. Because $E(\mathbb{F}_q)$ is a finite abelian group, it has a particularly simple form:

Theorem 2.2. *For an elliptic curve $E/\mathbb{F}_q$, there are positive integers N_1, N_2 with $N_2 \mid N_1$ and $N_2 \mid (q-1)$ such that*

$$E(\mathbb{F}_q) \simeq \mathbb{Z}/N_1\mathbb{Z} \times \mathbb{Z}/N_2\mathbb{Z}.$$

For an elliptic curve E/k and any positive integer N, the N-torsion subgroup is

$$E[N] = \{P \in E(\bar{k}) : [N]P = O\}.$$

If $\mathrm{char}(k)$ does not divide N, then

$$E[N] \simeq \mathbb{Z}/N\mathbb{Z} \times \mathbb{Z}/N\mathbb{Z}.$$

In characteristic p, the p-power torsion is more subtle. For $n \geq 1$, over an algebraic closure, as we will see later that $E[p^n]$ is trivial when E is *supersingular* and is isomorphic to $\mathbb{Z}/p^n\mathbb{Z}$ when E is *ordinary*.

✎ **Exercise 2.3.** Using Sage, compute the point sets and determine the group structures of the following curves over $\mathbb{F}_{23}$:

$$y^2 = x^3 + x + 1, \qquad y^2 = x^3 - x, \qquad y^2 = x^3 + 5x + 15.$$

✎ **Exercise 2.4.** Let $p > 3$, and let

$$E : y^2 = x^3 + ax + b, \qquad \widetilde{E} : y^2 = x^3 + ax - b$$

be elliptic curves over $\mathbb{F}_p$. Show that if $p \equiv 1 \pmod 4$, then $\#E(\mathbb{F}_p) = \#\widetilde{E}(\mathbb{F}_p)$; if $p \equiv 3 \pmod 4$, show that $\#E(\mathbb{F}_p) + \#\widetilde{E}(\mathbb{F}_p) = 2p + 2$. Hint: determine when -1 is a square in $\mathbb{F}_p$.

2.3 The discrete logarithm problem

Now we are ready to study the discrete logarithm problem, a problem that is easy to state, but believed to be hard to solve. The problem is defined in any finite group, but we will focus on the case of elliptic curves over finite fields.

Problem 2.5 (Elliptic-curve discrete logarithm problem, ECDLP)**.** Given an elliptic curve E over a finite field $\mathbb{F}_q$, a point $P \in E(\mathbb{F}_q)$ of prime order n, and a point $Q \in \langle P \rangle$, find the unique $d \in \{0, \ldots, n-1\}$ such that

$$Q = [d]P.$$

The integer d is called the discrete logarithm of Q to the base P.

Example 2.6. On the curve of Example 1.14, $P = (0, 1)$ has order 5 and $[3]P = (2, 2)$. Hence the discrete logarithm of $(2, 2)$ to the base P is 3. This tiny example is deliberately insecure: trying all five possibilities is easy.

✎ **Exercise 2.7.** Here is Sage Math code to create a larger example. Can you find the discrete logarithm of Q to the base P?

```
p = Integer(
    "115792089237316195423570985008687907853269984665640"
    "564039457584007908834671663"
)
F = GF(p)
E = EllipticCurve(F, [0, 0, 0, 0, 7])
P = E(
    Integer(
        "550662630222773436695787188951685343262506034537775941755001"
        "873603891167292400"
    ),
    Integer(
        "326705100207588169780830851305070431844712733806592432759389"
        "04335757337482424"
    )
)
Q = E(
    Integer(
        "289214210243673324786884154681051689448148248437866371452624650"
        "2832759657938"
    ),
    Integer(
        "944804112252944922688775491098289217072623569625897683169341750"
        "98490464915373"
    )
)
```

The best general classical attacks, such as Pollard rho, use roughly $\sqrt{n}$ group operations and little memory. Therefore a subgroup of order about 2^{256} gives an approximately 128-bit classical security target. Special curves or poorly chosen parameters can be weaker, so practical ECC uses standardized, audited curves and algorithms.

✎ **Exercise 2.8.** Let P have order n. Show that $[a]P = [b]P$ if and only if $a \equiv b \pmod{n}$. Why does this prove uniqueness in the ECDLP definition?

✎ **Exercise 2.9.** On the curve and with the generator $P = (0, 1)$ from Example 1.14, write down the table

$$O,\ P,\ [2]P,\ [3]P,\ [4]P.$$

Use it to find the discrete logarithm of each nonidentity point to the base P. Which part of this method becomes infeasible when the subgroup order is about 2^{256}?

✎ **Exercise 2.10** (Transporting a discrete logarithm)**.** Let $\varphi : E \to E'$ be a group homomorphism, and suppose that P has prime order n and that $\varphi(P)$ also has order n. If $Q = [d]P$, prove that

$$\varphi(Q) = [d]\varphi(P).$$

Show that a solution to the discrete logarithm problem for $(\varphi(P), \varphi(Q))$ recovers d modulo n. Why is the assumption on the order of $\varphi(P)$ essential?

2.4 Diffie–Hellman key agreement

Diffie–Hellman is a *key-agreement* protocol, not an encryption protocol. It lets two parties derive the same shared secret over a public channel in the chosen subgroup.

1. Alice and Bob agree on public parameters $(E, \mathbb{F}_q, P, n)$, where P has prime order n.
2. Alice samples $a \in \{1, \ldots, n-1\}$ uniformly and sends $A = [a]P$.
3. Bob samples $b \in \{1, \ldots, n-1\}$ uniformly and sends $B = [b]P$.
4. Alice computes $S_A = [a]B$ and Bob computes $S_B = [b]A$.

Both obtain the same point

$$S_A = [a]([b]P) = [ab]P = [b]([a]P) = S_B.$$

The protocol uses a key-derivation function (KDF) on an agreed encoding of S_A, together with the public transcript, to obtain symmetric key material; the point P is not used directly as an encryption key.

Example 2.11 (A toy ECDH exchange)**.** On the curve in Example 1.14, take the generator $P = (0, 1)$ of order 5. If Alice chooses $a = 2$ and Bob chooses $b = 3$, then

$$A = [2]P = (2, 5), \qquad B = [3]P = (2, 2).$$

Alice computes $[2]B = [6]P = P$, while Bob computes $[3]A = [6]P = P$. Thus they obtain the same shared point. The calculation also makes clear why the scalars are read modulo 5: here $6 \equiv 1 \pmod 5$.

Remark 2.12. One might try to imitate the protocol with a public nonzero integer p: Alice sends $A = ap$ and Bob sends $B = bp$, and they compute

$$aB = bA = abp.$$

This gives no secrecy. An eavesdropper who knows p, A, and B can recover $a = A/p$ and $b = B/p$, or compute the shared value directly as AB/p. Thus multiplication by a public integer is efficiently reversible by division. In elliptic-curve Diffie–Hellman, by contrast, recovering a from P and $[a]P$ is the presumed-hard discrete-logarithm problem.

✎ **Exercise 2.13.** Suppose an implementation accepts the point $B = O$ as Bob's public message. What shared point does Alice compute, regardless of her secret scalar a? Explain why checking subgroup membership alone does not rule out this case, and state an additional check the implementation should make.

✎ **Exercise 2.14.** In the subgroup generated by the point P of Example 1.14, let Alice choose $a = 2$ and Bob choose $b = 3$. Compute A, B, and the shared point.

✎ **Exercise 2.15.** Suppose that a generic discrete-logarithm attack requires approximately $\sqrt{n}$ group operations in a subgroup of prime order n.

1. Estimate the work factor when n is about 2^{256}.
2. If an attacker can carry out 2^{40} group operations per second, estimate, in years, how long 2^{128} operations would take.
3. Explain why this estimate motivates describing such a subgroup as offering roughly 128 bits of classical security.

Warning 2.16. Unauthenticated Diffie–Hellman does *not* authenticate Alice or Bob. An active attacker can replace A and B and mount a man-in-the-middle attack. Real protocols authenticate the exchange, for example with signatures, certificates, or a pre-shared authentication mechanism. Implementations must also validate received public points and handle cofactors according to the chosen curve protocol.

✎ **Exercise 2.17.** Alice sends $A = [a]P$ and Bob sends $B = [b]P$, but an active adversary Eve replaces Alice's message by $A' = [e_A]P$ and Bob's message by $B' = [e_B]P$, for scalars chosen by Eve.

1. Determine the point computed by Alice and the point computed by Bob.
2. Determine the two points Eve can compute from A, B, e_A, and e_B.
3. Explain how Eve can use these two shared points to read and alter messages if Alice and Bob do not authenticate the exchange.

2.5 What is assumed to be hard?

The computational Diffie–Hellman problem (CDH) in the subgroup $\langle P \rangle$ is: given

$$A = [a]P \qquad \text{and} \qquad B = [b]P,$$

compute $[ab]P$. The security of unauthenticated ECDH against a passive eavesdropper is modeled by the difficulty of this problem.

An algorithm for ECDLP solves CDH: recover a from (P, A) and then compute $[a]B$. The converse implication is not known in complete generality, so the two assumptions should not be identified. For well-chosen elliptic-curve groups, no general classical algorithm is known that is asymptotically faster than generic square-root search.

One curve that is widely used is the one recommended by the Internet Research Task Force (IRTF) given by the following equation over $\mathbb{F}_p$ with $p = 2^{255} - 19$:

$$E : y^2 = x^3 + 486662x^2 + x.$$

The base point recommended for use has coordinates $x = 9$ and $y =$

$$14781619447589544791020593568409986887264606134616475288964881837755586237401.$$

The order of this point is a prime number

$$n = 2^{252} + 27742317777372353535851937790883648493.$$

Moreover, NIST, the US National Institute of Standards and Technology, has a list of 15 elliptic curves that are recommended for use in cryptography. The curves are defined over prime fields $\mathbb{F}_p$ and binary fields $\mathbb{F}_{2^m}$, and have been chosen to have a prime number of points or a small cofactor.

Lecture 3: Isogenies of elliptic curves

The Diffie–Hellman key-agreement protocol studied in the previous lecture is not resistant to attacks by sufficiently powerful fault-tolerant quantum computers: Shor's algorithm can solve the discrete logarithm problem on which it relies. This motivates the search for new post-quantum public-key encryption and key-agreement protocols. Isogenies provide one mathematical setting in which such protocols have been explored, and this lecture develops the basic language needed to study them.

The definitions in this section are motivated from the point arithmetic already developed, but some proofs use algebraic geometry and are stated without proof. The important new idea is that an isogeny is a structure-preserving map from one elliptic curve to another.

3.1 Maps between elliptic curves

Definition 3.1. Let E and E' be elliptic curves over a field $\mathbb{F}_p$. In these notes, a map between E and E' is a function $\varphi : E \to E'$ given by formulas

$$\varphi(x, y) = \big(f(x, y), g(x, y)\big),$$

where f and g are quotients of polynomials in x and y. Away from zeros of their denominators, the formulas must send points of E to points of E', and the map sends the point at infinity O to O.

Example 3.2. Consider the curves $E : y^2 = x^3 - x$ and $E' : y^2 = x^3 + 4x$ over $\mathbb{F}_7$. For $x \neq 0$, put

$$\varphi(x, y) = \left(x - \frac{1}{x},\ y\left(1 + \frac{1}{x^2}\right)\right),$$

and set $\varphi(O) = \varphi(0, 0) = O$. If $(x, y) \in E$ and $x \neq 0$, then

$$\left(y\left(1 + \frac{1}{x^2}\right)\right)^2 = \left(x - \frac{1}{x}\right)^3 + 4\left(x - \frac{1}{x}\right),$$

so $\varphi(x, y)$ lies on E'. Thus these formulas define a map $E \to E'$. For instance, $(4, 2) \in E(\mathbb{F}_7)$ is sent to $(2, 3) \in E'(\mathbb{F}_7)$. We will recover this map from Vélu's formulas in Example 3.12.

Example 3.3. An example of something that is not a map is the following. Let $E : y^2 = x^3 + x$ and $E' : y^2 = x^3 + 2x$ over $\mathbb{F}_7$. Consider the formulas

$$\varphi(x, y) = (2x, y).$$

This is not a map because it does not send points of E to points of E'. For instance, $(1, 3) \in E(\mathbb{F}_7)$ is sent to $(2, 3)$, which is not a point of $E'(\mathbb{F}_7)$.

3.1.1 Isomorphisms and the j-invariant

Definition 3.4. Let $E : y^2 = x^3 + ax + b$ and $E' : y^2 = x^3 + a'x + b'$ be short Weierstrass curves over $\mathbb{F}_p$. They are isomorphic over $\mathbb{F}_p$ if there is a $u \in \mathbb{F}_p^\times$ such that

$$a' = u^4 a, \qquad b' = u^6 b.$$

The isomorphism sends (x, y) to $(u^2 x, u^3 y)$ and sends O to O.

Definition 3.5. The j-invariant of E is

$$j(E) = 1728 \frac{4a^3}{4a^3 + 27b^2} = \frac{256 \cdot 27a^3}{4a^3 + 27b^2}.$$

It is unchanged by isomorphism. Over an algebraic closure, two elliptic curves of characteristic different from 2 and 3 are isomorphic exactly when their j-invariants agree.

Example 3.6. Over $\mathbb{F}_7$, the curves $y^2 = x^3 + x$ and $y^2 = x^3 + 2x$ are isomorphic: take $u = 2$, for which $u^4 = 2$ in $\mathbb{F}_7$. The map is $(x, y) \mapsto (4x, y)$.

✎ **Exercise 3.7.** Verify directly that $(x, y) \mapsto (4x, y)$ sends points of $y^2 = x^3 + x$ over $\mathbb{F}_7$ to points of $y^2 = x^3 + 2x$. Find its inverse map, and compute the common j-invariant in $\mathbb{F}_7$.

✎ **Exercise 3.8.** Decide if the following statement is true or false: "Every isomorphism of elliptic curves is also a group isomorphism".

3.2 Isogenies

Isomorphisms identify curves that have exactly the same elliptic-curve structure. We now broaden the comparison: an isogeny is a map between possibly different curves that preserves the group law, while allowing a finite kernel.

Definition 3.9. Let E and E' be elliptic curves over a field $\mathbb{F}_p$. An isogeny is a nonconstant such map of elliptic curves defined over $\mathbb{F}_p$ that respects the elliptic-curve group law:

$$\varphi(P + Q) = \varphi(P) + \varphi(Q).$$

Its kernel is $\ker \varphi = \{P \in E(\overline{\mathbb{F}}_p) : \varphi(P) = O\}$, where $\overline{\mathbb{F}}_p$ denotes an algebraic closure of $\mathbb{F}_p$. The degree of φ is the size of its kernel, which is finite.

For short Weierstrass models (in characteristic different from 2 and 3), write

$$E \colon y^2 = x^3 + ax + b, \qquad E' \colon {y'}^2 = {x'}^3 + a'x' + b'.$$

Away from the finite set of points where a formula has a pole, an isogeny can be written in coordinates as

$$\varphi(x,y) = \big(r(x),\, y\,s(x)\big)$$

for rational functions $r(x), s(x) \in k(x)$, with $\varphi(O) = O$. The functions are not arbitrary: substituting into the equation of E' gives the identity

$$s(x)^2\big(x^3 + ax + b\big) = r(x)^3 + a'r(x) + b'.$$

Thus the coordinate formula sends points of E to points of E'. The additional group-law condition says that it also sends sums of points to sums of their images.

Example 3.10. Let $E\colon y^2 = x^3 - x$ defined over a prime field k of characteristic different from 2 and 3. When $y \neq 0$, we have the map $[2] : E \to E$ that sends P to $[2]P$. In coordinates, it is

$$[2](x,y) = (r(x),\ y\,s(x))\,,$$

where

$$r(x) = \frac{x^4 - 2ax^2 - 8bx + a^2}{4(x^3 + ax + b)}, \qquad s(x) = -1 + \frac{(3x^2 + a)(x - r(x))}{2(x^3 + ax + b)} \in k(x).$$

Thus $[2]$ is algebraic: on this open set its coordinates are rational functions in x and y. Multiplication by 2 sends O to O. It is nonconstant because the numerator of $r(x)$ has degree 4, while its denominator has degree 3; hence $r(x)$ cannot be a constant rational function.

The map $[2] : E \to E$ is therefore an isogeny. Its degree corresponds to the number of points over $\bar{k}$ such that $[2]P = P$, so the 2-torsion of E. We can compute that the points $(0,0), (1,0), (-1,0)$ are of 2-torision. Since $\#E[2] \leq 4$, then we are done and the kernel of $[2]$ is $E[2]$. Then, the isogeny $[2]$ is of degree 4.

Besides O, the points in this kernel are the three points with $y = 0$ over $\bar{k}$; their x-coordinates are the roots of $x^3 + ax + b$. Even if none of those roots is in k, the kernel still has four points over $\bar{k}$. This is why the definition of a kernel uses an algebraic closure.

3.2.1 Subgroups and isogenies

Theorem 3.11. *If $G \subseteq E(\bar{k})$ is a finite subgroup stable under the relevant field automorphisms, then there is an elliptic curve E/G and an isogeny*

$$\pi_G : E \longrightarrow E/G$$

with kernel G. In explicit computations, Vélu's formulas construct E/G and π_G from the points of G.

Vélu's formulas make the preceding theorem explicit. Suppose that $\mathrm{char}(k) \neq 2, 3$, that

$$E\colon y^2 = x^3 + ax + b,$$

and let $G \subseteq E(\bar{k})$ be a finite subgroup stable under the relevant field automorphisms. Put $G^* = G \setminus \{O\}$. Then a short Weierstrass model for the quotient is

$$E/G\colon Y^2 = X^3 + a_G X + b_G,$$

where

$$a_G = a - 5 \sum_{Q \in G^*} \bigl(3x(Q)^2 + a\bigr),$$
$$b_G = b - 7 \sum_{Q \in G^*} \bigl(5x(Q)^3 + 3a\, x(Q) + 2b\bigr).$$

For an affine point P for which the expressions are finite, the quotient map is given by

$$\pi_G(P) = \bigl(X(P), Y(P)\bigr), \qquad \begin{aligned} X(P) &= \sum_{Q \in G} x(P+Q) - \sum_{Q \in G^*} x(Q), \\ Y(P) &= \sum_{Q \in G} y(P+Q). \end{aligned}$$

These rational expressions extend to all of E, with every point of G mapping to O. Thus, given the points of the kernel, one can compute both the neighboring curve and the isogeny to it; see Vélu [V́71].

Example 3.12. Let $E\colon y^2 = x^3 - x$ over $\mathbb{F}_7$. Its nonidentity 2-torsion points are $(0,0)$, $(1,0)$, and $(6,0)$. Take $G = \langle (0,0) \rangle$, so that $G^* = \{Q\}$ with $Q = (0,0)$. Here $a = -1$ and $b = 0$. Vélu's coefficient formulas give

$$a_G = -1 - 5\bigl(3 \cdot 0^2 - 1\bigr) = 4, \qquad b_G = 0 - 7\bigl(5 \cdot 0^3 + 3(-1) \cdot 0 + 2 \cdot 0\bigr) = 0 \quad \text{in } \mathbb{F}_7.$$

Thus the quotient curve computed by Vélu's formulas is

$$E/G\colon y^2 = x^3 + 4x.$$

Now let $P = (x, y)$ with $x \neq 0$. The addition formula with $Q = (0,0)$ gives

$$x(P+Q) = \left(\frac{y}{x}\right)^2 - x = -\frac{1}{x}, \qquad y(P+Q) = \frac{y}{x}\left(x + \frac{1}{x}\right) - y = \frac{y}{x^2}.$$

Since $x(Q) = y(Q) = 0$, Vélu's coordinate formulas therefore give

$$\pi_G(x,y) = \bigl(x + x(P+Q),\ y + y(P+Q)\bigr) = \left(x - \frac{1}{x},\ y\left(1 + \frac{1}{x^2}\right)\right).$$

The formulas extend with $\pi_G(O) = \pi_G(0,0) = O$. This formula is visibly nonconstant, and direct substitution gives

$$y^2\left(1 + \frac{1}{x^2}\right)^2 = \left(x - \frac{1}{x}\right)^3 + 4\left(x - \frac{1}{x}\right),$$

so it sends points of E to points of E/G. Vélu's construction shows that it preserves the group law; hence it is a 2-isogeny with kernel $\{O, (0,0)\}$. For instance,

$$\pi_G(4,2) = (2,3).$$

Remark 3.13. You can define isogenies in Sage as quotients. For example, the isogeny in Example 3.12 can be computed as follows:

```
F = GF(7)
E = EllipticCurve(F, [0, 0, 0, -1, 0])
P = E(0, 0); Q = E(1, 0)     # Generators of 2-torsion
phi = E.isogeny([P, Q])
```

The output is the isogeny π_G with kernel $\langle P, Q\rangle$. You can evaluate it on points of E and check that it sends them to points of the codomain curve. To find the codomain curve, you can use `phi.codomain()`.

✎ **Exercise 3.14.** Let P have order ℓ. Why is $\langle P\rangle$ an allowable candidate for the kernel of an ℓ-isogeny when it is defined over the base field? What information must Vélu's formulas take as input?

✎ **Exercise 3.15.** Find all curves over $\mathbb{F}_7$ that are isogenous to $y^2 = x^3 - x$. For each, also find the degree of the isogeny.

Lecture 4: Isogeny graphs

Now we are ready to study isogeny graphs, which are the main objects of interest in isogeny-based cryptography. We will see that they have a rich arithmetic structure and that they can be used to construct cryptographic protocols that are believed to be secure against quantum attacks.

4.1 More about isogenies

We have already seen that isogenies are maps between elliptic curves that respect the group law. We now introduce a special class of isogenies that will be particularly useful in cryptography.

Definition 4.1. If ℓ is a prime different from the characteristic of $\mathbb{F}_q$, an ℓ-isogeny is an isogeny whose kernel

$$\ker \varphi = \{P \in E(\overline{\mathbb{F}}_q) : \varphi(P) = O\}$$

has exactly ℓ points. This is equivalent to saying that φ has degree ℓ.

Example 4.2. Consider the multiplication-by-2 isogeny $\varphi\colon E \to E$ from Example 3.10, where

$$E : y^2 = x^3 - x.$$

A point belongs to $\ker \varphi$ exactly when doubling it gives O; in other words, it has order dividing 2. Besides O, these are precisely the points with $y = 0$. Since

$$x^3 - x = x(x-1)(x+1),$$

we obtain

$$\ker \varphi = \{O, (0,0), (1,0), (-1,0)\}.$$

The kernel has four points, so φ has degree 4. In particular, multiplication by 2 is *not* a 2-isogeny: the 2 in φ describes the group operation, whereas a 2-isogeny has a kernel of size 2.

Example 4.3. Over $\mathbb{F}_7$, consider the curves

$$E : y^2 = x^3 - x, \qquad E_1 : y^2 = x^3 + 4x.$$

Example 3.12 constructs an isogeny $\varphi_1 : E \to E_1$ given, for $x \neq 0$, by

$$\varphi_1(x,y) = \left(x - \frac{1}{x},\ y\left(1 + \frac{1}{x^2}\right)\right),$$

together with $\varphi_1(O) = \varphi_1(0,0) = O$.

Note that no point with $x \neq 0$ is sent to O. Hence

$$\ker \varphi_1 = \{O, (0,0)\},$$

and φ_1 is a 2-isogeny.

✎ **Exercise 4.4.** Here are two more isogenies of the elliptic curve E from Example 4.3 (you do not need to prove that they are isogenies). Show that they both have degree 2.

$$\begin{array}{rccc} \varphi_2\colon & E & \to & E_2\colon y^2 = x^3 + 3x \\ & (x,y) & \mapsto & \left(\dfrac{x^2+x+2}{x+1}, \dfrac{x^2y+2xy-y}{x^2+2x+1}\right) \end{array}$$

$$\begin{array}{rccc} \varphi_3\colon & E & \to & E_2\colon y^2 = x^3 + 3x \\ & (x,y) & \mapsto & \left(\dfrac{x^2-x+2}{x-1}, \dfrac{x^2y-2xy-y}{x^2-2x+1}\right) \end{array}$$

Remark 4.5. It turns out that the isogenies from Example 4.3 and Exercise 4.4 are all of the 2-isogenies of E defined over $\mathbb{F}_7$.

✎ **Exercise 4.6.** Let E be an elliptic curve defined over $\mathbb{F}_p$ and φ be an isogeny of E of degree ℓ. If n is a positive integer coprime to ℓ, show that the restriction of φ to a subgroup of order n is injective.

4.2 Ordinary and supersingular elliptic curves

Not all curves are equally useful for isogeny-based constructions. The following definition distinguishes two important classes of curves in characteristic p.

Definition 4.7. For an elliptic curve E in characteristic p, the p-torsion is

$$E[p] = \{P \in E(\bar{\mathbb{F}}_p) : [p]P = O\}.$$

The curve is ordinary if $E[p] \simeq \mathbb{Z}/p\mathbb{Z}$, and supersingular if $E[p] = \{O\}$.

The word "supersingular" is a property in characteristic p, not a claim that the curve equation is singular. Every curve considered here remains nonsingular. Supersingular curves are central to several isogeny-based constructions because their endomorphisms and isogeny graphs have unusually rigid structure.

Example 4.8. Let's consider the elliptic curve $E\colon y^2 = x^3 - x$ over $\mathbb{F}_7$. We have computed that $\#E(\mathbb{F}_7) = 8$, so $E[p]$ cannot be isomorphic to $\mathbb{Z}/p\mathbb{Z}$. Then, E is supersingular.

Example 4.9. Over $\mathbb{F}_5$, direct point counting gives

$$\#\{y^2 = x^3 + 1\}(\mathbb{F}_5) = 6, \qquad \#\{y^2 = x^3 + x\}(\mathbb{F}_5) = 4.$$

The first curve is supersingular and the second is ordinary. Their Frobenius traces $a_5 = 5 + 1 - \#E(\mathbb{F}_5)$ are respectively 0 and 2. These examples illustrate that ordinary and supersingular refer to characteristic-p torsion, not to whether the curve has a singular point; see [Sil09, Ch. V, §4].

✎ **Exercise 4.10.** Make the square table in $\mathbb{F}_5$ and verify the two point counts in the preceding example. Which curve has $j = 0$, and which has $j = 1728$?

Theorem 4.11 (Tate)**.** *Two elliptic curves over $\mathbb{F}_q$ are $\mathbb{F}_q$-isogenous if and only if they have the same number of $\mathbb{F}_q$-points.*

The proof is beyond the scope of this course, but we can check an example. Try the following SageMath code to see which curves over $\mathbb{F}_7$ are isogenous to $y^2 = x^3 + 1$.

```
F = GF(7)

E = EllipticCurve(F, [0, 0, 0, 0, 1])   # y^2 = x^3 + 1

for a in F:
    for b in F:
        # y^2 = x^3 + a*x + b must be nonsingular
        if 4*a^3 + 27*b^2 == 0:
            continue

        Eab = EllipticCurve(F, [0, 0, 0, a, b])

        if Eab.is_isogenous(E):
            print("Isogenous: a =", a, ", b =", b)
        else:
            print("Not isogenous: a =", a, ", b =", b)

        print(Eab.cardinality(), "points in Eab;",
            E.cardinality(), "points in E")
        print()
```

All supersingular elliptic curves over $\bar{\mathbb{F}}_p$ belong to one isogeny class. This does *not* mean that every two vertices are joined by a single ℓ-isogeny: generally they are connected by a path whose total degree is a power of ℓ.

Exercise 4.12. The following elliptic curves are all defined over $\mathbb{F}_7$. Determine which are ordinary and which are supersingular.

$$\begin{aligned} E_1 &: y^2 = x^3 + 1, & E_2 &: y^2 = x^3 + x + 2, \\ E_3 &: y^2 = x^3 + 6x, & E_4 &: y^2 = x^3 + 2x + 1, \\ E_5 &: y^2 = x^3 + 3, & E_6 &: y^2 = x^3 + 4x + 4. \end{aligned}$$

It might help you to compute the number of points on each curve for which the Sage command is `E.cardinality()` and to check for isogenies with the command `E.is_isogenous(E2)`.

4.3 Isogeny graphs

For cryptography, we need a setting in which individual operations are easy to compute but recovering hidden information is difficult. A small ℓ-isogeny can be computed from its kernel, so it is natural to regard it as one efficient step from its source curve to its target curve. A secret can then be represented by a sequence of such steps, while an adversary sees only the initial and final curves. Recording all possible small steps at once gives an isogeny graph.

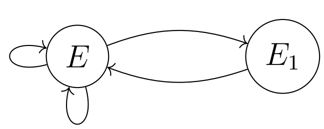
Definition 4.13. Fix p, a prime $\ell \neq p$, and a collection $\mathcal{V}$ of elliptic curves over $\mathbb{F}_{p^2}$, considered up to isomorphism and closed under the quotients below. The ℓ-isogeny graph has one vertex for each isomorphism class in $\mathcal{V}$. For every subgroup $C \subset E(\overline{\mathbb{F}}_p)$ of order ℓ, it has a directed edge

$$E \longrightarrow E/C.$$

Thus edges record the quotient ℓ-isogenies by their kernels. Multiple edges and loops are retained.

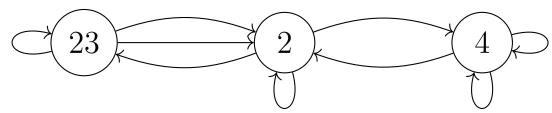
When convenient, a vertex is labelled by its j-invariant. This is only a label: the definition is in terms of isomorphism classes and quotient isogenies. Since $E[\ell] \simeq (\mathbb{Z}/\ell\mathbb{Z})^2$, every vertex has $\ell + 1$ outgoing edges in this convention. The choice of $\mathcal{V}$ matters. Ordinary components have volcano-like structure, while the usual supersingular ℓ-isogeny graphs form a very different family.

Example 4.14. Let $E : y^2 = x^3 - x$ over $\mathbb{F}_7$. This time, take vertices up to $\mathbb{F}_7$-isomorphism and draw only 2-isogenies defined over $\mathbb{F}_7$. By Example 4.8, E is supersingular. Example 4.3 and Exercise 4.4 give its three 2-isogenies. The target $E_2 : y^2 = x^3 + 3x$ is $\mathbb{F}_7$-isomorphic to E, whereas $E_1 : y^2 = x^3 + 4x$ is not. Thus φ_2 and φ_3 give two loops at the vertex E, while φ_1 gives an edge from E to E_1. The 2-isogeny from E_1 with kernel $\{O, (0, 0)\}$ has target $y^2 = x^3 + 5x$, which is $\mathbb{F}_7$-isomorphic to E. The resulting $\mathbb{F}_7$-rational graph is therefore



The two loops have kernels generated by $(1,0)$ and $(-1,0)$; the two directed edges between the vertices have kernel generated by $(0,0)$ on their respective source curves.

Example 4.15. For $p = 31$ and $\ell = 2$, the supersingular 2-isogeny multigraph has three vertices with j-invariant labels 23, 2, and 4. Parallel edges and loops record distinct isogenies, so they should not be erased when drawing the graph.



✎ **Exercise 4.16.** In the displayed $p = 31$, $\ell = 2$ multigraph, count the incoming and outgoing edges at each vertex, retaining loops and parallel edges. Verify that each vertex has three outgoing edges, as suggested by $\ell + 1 = 3$. Are the incoming edge counts the same in this directed drawing? Explain why direction and multiplicity must be specified before making a regularity statement.

After vertices are identified up to isomorphism, exceptional automorphisms can make the incoming counts differ from the outgoing counts, as in the example. Standard treatments use suitable weights (or explicitly account for the exceptional vertices) to obtain the regular graph whose expansion is studied. A k-regular Ramanujan graph has every *nontrivial* adjacency eigenvalue λ satisfying

$$|\lambda| \leq 2\sqrt{k-1}.$$

The trivial eigenvalue is k, so it must be excluded from this inequality. The precise graph, multiplicities, and treatment of exceptional automorphisms matter in the theorem; see [DFJP14].

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